



Predicting water quality index using machine learning techniques: a case study of river Ganga in Haridwar, India

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Received: 8 December 2024 / Accepted: 21 March 2025 / Published online: 9 April 2025
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Abstract

Water pollution presents considerable challenges to ecosystems, human health, and economic stability, underscoring the importance of robust monitoring and management systems. Water quality monitoring and management remain critical global challenges, particularly in achieving UN Sustainable Development Goal (SDG) 6. This research leverages Machine Learning (ML) techniques to predict the Water Quality Index (WQI) for Haridwar, a region in North India, utilising a comprehensive dataset that includes parameters like pH, Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), and ion concentrations. Multiple ML models were employed, including Support Vector Regression (SVR), Random Forest Regressor (RFR), Extreme Gradient Boosting Regressor (XGBR), and Recurrent Neural Network (RNN). The models were evaluated using various performance metrics; the RNN model demonstrated superior performance, achieving an RMSE of 0.7535, MAE of 0.5742, MSE of 0.3004, and an R^2 score of 0.9655. The WQI for the various sites considered ranged between 90 and 110, with an average WQI of 96 indicating moderate water quality conditions, revealing seasonal fluctuations, and periods of moderate pollution indicating potential water quality concerns. These results emphasize the effectiveness and precision of ML-based methods for water quality prediction, providing a valuable tool for proactive environmental management. This study highlights the potential of such models in promoting sustainable river ecosystem management, aligning with the objectives of SDGs 6 and 11. Furthermore, the methodological approach is versatile and can be adapted for water quality assessments in various geographical settings.

Keywords Water quality management · Environmental monitoring · Sustainable development · Machine learning

Introduction

Water is essential to all living things and human society; it is the foundation of ecosystem health and sustainable development. Even while water quality is necessary, it is declining at

a startling rate globally (Jat Baloch et al. 2022). According to (UN Water 2023), 2 billion people worldwide lack access to safely managed drinking water services, while roughly 80% of wastewater returns to ecosystems untreated. This scenario is especially acute in emerging countries where industrialization and urbanization frequently accelerate faster than infrastructural development (Ayejoto et al. 2023).

The Ganga, the holiest river in India, is a prime example of this worldwide problem with water quality. It is more than 2,525 km long, and provides for the livelihood of more than 400 million people, and has significant cultural and religious value (Chaturvedi 2012). Haridwar is an important study site for several reasons. It is located in Uttarakhand, where the Ganga flows from the Himalayas into the Indo-Gangetic plains. Millions of pilgrims visit here yearly for religious rites and ceremonial bathing, making it one of the seven holiest sites in Hindu tradition. An unusual combination of factors affecting water quality is brought about by its cultural significance, increasing urbanization, and industrial expansion (Trivedi 2010).

Communicated by: Hassan Babaie

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Haridwar's water quality about the environment is a complex problem. Untreated sewage outflow, industrial effluents from nearby industrial estates, agricultural runoff, and religious activities are the primary sources of contamination (Matta et al. 2020). Studies conducted by the Central Pollution Control Board of India, particularly during religious festivals, have shown increased fecal coliform counts and biological oxygen demand (BOD). These contaminants significantly impact residents, affecting irrigation methods, the region's abundant aquatic biodiversity, and drinking water quality. Increasing industrialization and urbanization result in a unique combination of factors affecting water quality (Matta et al. 2021). Several states provide the local populace with drinking water, bathing facilities, fishing, and agricultural services along the Ganga River. Serving an estimated 500 million people, the Ganga is one of the most significant rivers in the world, but the discharge of industrial waste and untreated wastewater severely damages it. Several things, including the dense population, numerous shoreline celebrations, trash disposals, and corpse disposals, have contaminated the Ganga River (Jat Baloch et al. 2021). The river is treated with various chemicals, including insecticides, herbicides, and fertilizers. Using the Water Quality Index (WQI), the water quality in a particular area can be analyzed, and information concerning water quality can be provided in the most straightforward way possible to the general public, government officials, and others. The water quality information derived from raw data (Cabassud et al. 2001) is then compiled into logical and simplified forms by WQI, which divides it into categories based on its overall water quality.

WQIs help communicate water quality information to stakeholders and policymakers because they represent the composite influence of several water quality parameters (Chidiac et al. 2023). They are essential for accurately assessing and communicating water quality. The National Sanitation Foundation Water Quality Index (NSF-WQI) and the Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) are extensively used globally. One of the most widely used indices in the US and worldwide is the NSF-WQI. Integrating many water quality metrics into a single numerical score that reflects the overall health of a body of water simplifies the evaluation of water quality (Nigar et al. 2024). Specialized indices, such as the Industrial Water Quality Index (InWQI) and the Irrigation Water Quality Index (IWQI), offer focused evaluations for particular applications (Drew 2017). InWQI evaluates parameters like pH, TDS, hardness, and conductivity to determine water suitability for various industrial processes, with standards varying based on specific industry requirements and equipment sensitivity. The IWQI assesses irrigation water suitability by measuring electrical conductivity, sodium adsorption ratio, and specific ion concentrations to prevent soil degradation and ensure optimal crop

productivity. Additionally, the Bureau of Indian Standards (BIS) has created region-specific indexes considering regional water consumption trends and environmental factors.

Artificial Intelligence (AI) accurately predicts WQI with advanced algorithms and data analysis techniques (Hussein et al. 2023). Data sources like water quality sensors, satellite imagery, and environmental databases can be analyzed using AI-based algorithms like Machine Learning (ML) and Deep Learning (DL). Predicting water quality parameters accurately using these algorithms by identifying complex patterns and relationships in the data is possible. Researchers and environmental agencies can improve water quality monitoring and management accuracy and efficiency by incorporating AI into WQI prediction models. Predictive models powered by AI can help identify potential sources of contamination, forecast water quality changes, and detect water pollution events early (Agbasi et al. 2024).

Several studies have investigated these techniques to accurately predict WQI, including support vector machines, RFs, neural networks, and ensemble methods. A vast dataset containing parameters like pH, turbidity, dissolved oxygen, and various ion concentrations can be used to train these models (Egbueri and Agbasi 2022, Egbueri et al. 2023). According to research, ML models outperform traditional statistical methods in handling nonlinear relationships and interactions between water quality parameters. These models have also become more predictive due to improvements in feature selection and data preprocessing techniques (Singh et al. 2022).

Shakhari and Banerjee (2019) used a multi-class classification system based on a continuous data set to regulate water quality. A decision tree classification algorithm was used to forecast water quality and make data analysis effective. Water quality monitoring equipment is being researched to categorize drinking water quality from ponds, pipelines, and bottles. The quality of two classification and regression techniques was examined. According to Bisht et al. (2017), a River Water Quality Forecasting Model (RWQFM) based on Artificial Neural Networks (ANN) was used for monitoring the river Ganga, India. Five testing stations were chosen along the stream Ganga in Uttarakhand, India, to develop a model. The dataset comprised of data from 2001 to 2015, in which four water quality parameters were collected. Through several experiments with the Weka data mining tool, an optimal model was developed using one of the proficient ML approaches, which gave a very high degree of accuracy.

To forecast groundwater quality in the Shivganga River basin, which is situated on the eastern slopes of the Western Ghats, Kadam et al. (2019) employed ANN and Multiple Linear Regression (MLR) approaches. During and after the 2015 monsoon season, 34 groundwater samples were gathered and examined for the presence of significant cations

and anions. This study applied the Levenberg–Marquardt three-layer back propagation algorithm to ANN architectures to create a consistent and precise model for predicting groundwater quality. ANN prediction efficiency was also verified using the MLR model, which confirmed that the model produced consistently acceptable forecasts and consistently accurate performance for both seasons. Similarly, using Principal Component Analysis (PCA), the authors selected and weighted 12 water quality indicators from 19 sites for four rivers in Dhaka, Biriganga, Turag, Balu, and Shitalakhya (Roy et al. 2024).

To evaluate the water quality and learn more about Delhi's general water quality situation, Gupta et al. (2024) conducted an extensive study close to the Yamuna River and industry. Several water quality metrics and water samples from the Central Pollution Control Board (CPCB) from the previous eight years (2013–2021) were utilized to estimate WQI. The achieved accuracy of 93.6% was impressive. Bhardwaj and Singh (2022) used a multilayer perceptron (MLP) and radial basis function neural network model of DO using the ANN model, proving that it can recognize intricate nonlinear correlations between inputs and outputs for a study conducted in Uttar Pradesh. An examination of the models showed that a multi-layer perceptron model had a more substantial correlation coefficient (R^2 score = 0.993) and a lower mean square error (Root Mean Square Error (RMSE) = 0.1984).

A multiple linear regression model, an MLP model tree, an SVR model, and an RF regression model were employed in (Nouraki et al. 2021) to forecast TDS, SAR, and TH variables in the Karun River from 1999 to 2019. PCA was first used to lower the number of input variables. TH, sodium (Na), chloride (Cl), and Na and Cl have the most significant effects on TDS and SAR. The RFR model has the best prediction accuracy for TDS (R^2 score = 0.98, RMSE = 70.50 mg l⁻¹). Based on data from 2005 to 2020, (Malek et al. 2022) explored a series of ML models for predicting the Kelantan River's Water Quality Classification (WQC). Thirteen physical and chemical parameters were used in the proposed methodology. Seven ML models, such as Decision Trees (DT), ANN, K-Nearest Neighbor (KNN), Naïve Bayes (NB), Support Vector Machine (SVM), RF, and Gradient Boosting (GB), were used for classification. The ensemble model of GB with a learning rate of 0.1 had the highest accuracy (94.90%) and the best prediction performance compared to the other algorithms.

Twelve ML models, including five boosting algorithms, three DT, and four ANN algorithms, were evaluated for their performance in predicting the La Buong River's water quality in Vietnam (Khoi et al. 2022). From 2010 to 2017, four monitoring stations along the La Buong River were used to calculate the WQI. XGB performed the best (R^2 score = 0.989, RMSE = 0.107) out of all the techniques. In

Abba et al. (2020), three stations across the Yamuna River in India were investigated for predicting WQIs, namely Nizamuddin, Palla, and Udi (Chambal), using Back Propagation Neural Networks (BPNN) and Adaptive Neuro-Fuzzy Inference Systems (ANFIS), SVR, and MLRs. A nonlinear ensemble method was proposed to improve the accuracy of single models using Neural Network Ensembles (NNEs). RMSE values varied between 0.1213 and 0.4107, 0.003 and 0.0367, and 0.002 and 0.0272, respectively. An ML model was developed for WQI prediction of the Talaldanda Canal, Orissa, according to Bhoi et al. (2022). ML prediction models used the new canal water input parameters to establish the water quality class. During the project's training phase, the Odisha State Pollution Control Board gathered a variety of water quality measurements, such as pH, DO, total coliform, and biochemical oxygen demand (BOD). DT, neural network, KNN, NB, support vector machine, and RF were used in the study. DT outperforms other classification algorithms with 96.6% accuracy.

Using extra tree regression (ETR), a novel ensemble ML model was created to forecast monthly WQI values on the Lam Tsuen River in Hong Kong. Several conventional stand-alone models, such as SVR and decision tree regression (DTR), are evaluated compared to the ETR model. With an R^2 score of 0.98 and RMSE of 2.99 (Asadollah et al. 2021), the ETR model, created using a monthly input data set of ten parameters, generates more accurate WQI predictions throughout the training and testing stages. A predictive model for river water quality was used to calculate the index value based on the water quality standards (Nair and Vijaya 2022). 11 sampling stations were set up along the Bhavani River, which flows through Kerala and Tamil Nadu. Water quality is affected by 27 things, including dissolved oxygen, temperature, pH, alkalinity, hardness, chlorine, and coliforms. Building an ML model requires normalizing and selecting features. An algorithm was built to predict water quality using linear regression, multilevel regressors, support vector regressions, and RFs. SVMs, NB, DT, and MLP classifiers were used to classify WQIs. MLP regressors accurately predict WQI with an RMSE of 2.432, while MLP classifiers accurately classify WQI 81% of the time.

Despite significant advancements in water quality assessment using ML techniques, several research gaps persist. Many existing studies focus on isolated parameters or limited geographic areas, failing to capture the complex, multi-parameter interactions that influence WQI predictions across diverse ecosystems. Our work addresses these gaps by analyzing a comprehensive dataset from six key sites in Haridwar, encompassing diverse parameters and using advanced ML models for achieving high prediction accuracy.

This study is motivated by the urgent need to monitor and control the Ganga River's water quality, especially at Haridwar. This area is important from a cultural, religious, and

economic standpoint. The water quality has been severely harmed by urbanization and industrial discharge, endangering aquatic ecosystems, human health, and the river's general sustainability. Understanding the degrees of pollution and the variables influencing the Ganga River requires evaluating the water quality using essential indicators, including pH, dissolved oxygen (DO), dissolved oxygen demand (BOD), chemical oxygen demand (COD), nutrients, and heavy metals. The objectives of the proposed study are as follows:

- Develop a dataset by systematically converting and validating historical water quality measurements from the Uttarakhand Pollution Control Board (UPCB) website (<https://ueppcb.uk.gov.in/>), spanning 2017–2022, with standardized parameters and consistent units to enable computational analysis.
- Implement multiple machine learning models (including techniques such as Random Forest, XGBoost etc.) to generate high-accuracy forecasts of water quality parameters and Water Quality Index (WQI) values across six strategically selected monitoring sites in Haridwar
- Execute a comprehensive spatiotemporal analysis of water quality parameters by examining seasonal fluctuations, long-term trends, and parameter correlations across the monitoring network in Haridwar.
- Conduct an extensive comparative analysis of the developed ML models against existing water quality prediction methodologies through standardized performance metrics, emphasizing improvements in prediction accuracy.

This study addresses the critical need for advanced water quality prediction tools in religiously and culturally significant locations facing multiple environmental pressures. It aims to develop a robust predictive framework for the Ganga River in Haridwar by integrating traditional water quality assessment methods with modern machine learning techniques. This framework could serve as a model for similar historic riverine cities worldwide struggling to balance cultural heritage, urban development, and environmental protection.

Materials and methods

The methodology for predicting the Ganga River's WQI comprises several systematic phases. Water quality parameters were initially obtained from the Uttarakhand Pollution Control Board database (UPCB 2024). The acquired dataset underwent preprocessing, including data cleaning and imputation of missing values, to ensure robust WQI computation. Subsequently, machine learning (ML) models were developed and trained using a portion of the preprocessed dataset.

The models' predictive capabilities were then evaluated on a separate test dataset. Model performance was assessed using multiple evaluation metrics to determine the optimal approach for WQI prediction. Figure 1 presents a schematic representation of the methodological framework.

Study area

The study region, Haridwar, is a geographically and environmentally significant location in the northern Indian state of Uttarakhand. Geologically, it is located in the foothills of the Himalayas and is distinguished by sedimentary deposits from the Ganga River, which traverses the area. The climate of Haridwar is predominantly subtropical, with distinct seasons: hot summers, monsoon rains, and cool winters. The area's topography combines plains and rolling hills, with the Shivalik Hills providing a striking backdrop. At Haridwar, the Ganga River, which rises from the Gangotri Glacier between latitude 29°55'04.64"N – 29°57'05.69"N and longitude 78°09'39.95"E – 78°10'37.34"E gives way from the untamed Himalayan landscape to the vast Indo-Gangetic plains. This geographic setting influences the area's hydrology and water quality dynamics and makes it a critical location for religious, agricultural, and ecological activities.

The banks of the Ganga are frequented for religious and cultural purposes, and the river itself is revered in Hinduism as the goddess Ganga. Named the 'National River' of India since 2008, the Ganga River is home to thousands of villages, 97 towns, and 29 major cities (Singh et al. 2022). However, maintaining sustainable water resource management and safeguarding river ecosystems and communities depend on controlling and balancing river water usage. In addition to industrial discharges, agriculture, domestic waste, municipal and solid waste, mining and extractive activities, urban runoff, agricultural livestock operations, erosion, deforestation, and climate change, there are still several familiar sources of pollution. Figure 2 represents an image of the study area site for the Haridwar city for WQI prediction.

Data collection

Data was collected from the official website of the UPCB (<https://ueppcb.uk.gov.in/>) in which water samples were collected from six sampling sites, as depicted in Table 1, along with their Latitude (Ld) and Longitude (Lg). These monitoring stations were chosen based on their geographical distribution and proximity to major urban centers, industrial zones, and religious sites that potentially influence water quality.

This dataset contains monthly samples taken from these locations of the Ganga River between 2017 and 2022. It comprises 432 data points, representing monthly water

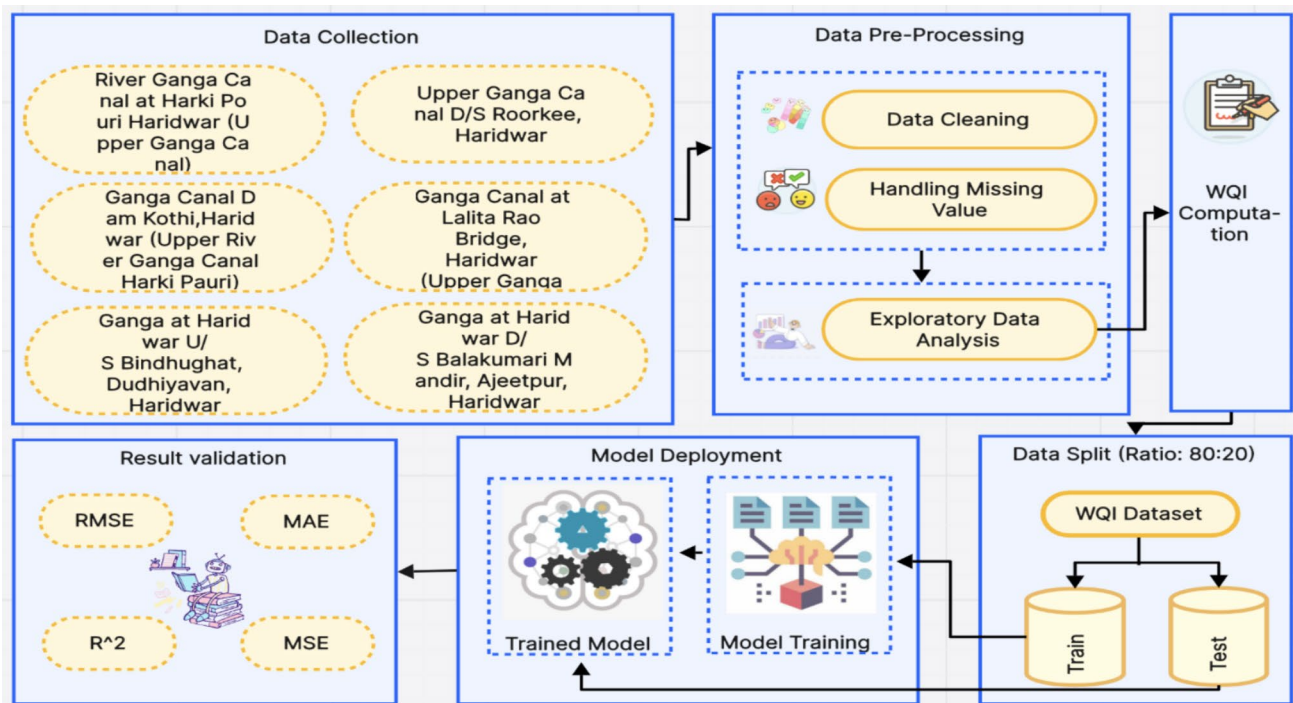


Fig. 1 Proposed Methodology Diagram



Fig. 2 Study Area of the site for Water Quality Prediction

Table 1 Data Collection Sites for WQI Prediction

Location	Location presented as	Coordinates
Upper Ganga Canal D/S Roorkee Haridwar	Site 1	Ld: 29.8540° N, Lg: 77.8880° E
Upper Ganga Canal D/S Harkipouri Haridwar	Site 2	Ld: 29.9467° N, Lg: 78.1622° E
Upper Ganga Canal D/S Harkipouri, Haridwar (Damkothi)	Site 3	Ld: 29.9503° N, Lg: 78.1576° E
Upper Ganga Canal Lalita Rao Bridge, Haridwar	Site 4	Ld: 29.9304° N, Lg: 78.1160° E
Upper Ganga Canal D/S Balkumari Mandir, Ajeetpur, Haridwar,	Site 5	Ld: 29.8895° N, Lg: 78.0481° E
Ganga U/S Bindughat Dudhiyabad, Haridwar,	Site 6	Ld: 29.9752° N, Lg: 78.1880° E

quality data collected from six sites along the Ganga River in Haridwar between 2017 and 2022. Water quality parameters, including temperature (°C), pH, DO, BOD (mg/L), total coliform (MPN/100 mL), alkalinity (mg/L), chlorides (mg/L), TDS (mg/L), calcium (mg/L), and hardness (mg/L), were measured in the samples.

Data pre-processing

When building ML models, it is imperative to ensure excellent data quality. By reducing data noise, pre-processing techniques increase the speed, generalization capacity, and execution rate of algorithms when dealing with large datasets. Selecting features, managing missing data, and treating outliers are essential when extracting and monitoring data. In the context of WQI prediction, extensive data pre-processing is crucial to improving the accuracy and reliability of the model by refining the dataset. When input data is adequately pre-processed—for example, by handling and standardizing missing values—it is better prepared for these algorithms (Panahi et al. 2022). Pre-processing enhances model performance by removing biases and outliers, thus creating more insightful data for accurate WQI predictions, which has been done in the study.

A crucial aspect of dataset pre-processing is managing missing values. The dataset used in this analysis has relatively little missing data (about 2%), primarily from measurements taken in October 2017 that were unavailable from all monitoring sites. This consistent pattern suggests that the missing data is limited to a particular temporal segment rather than randomly dispersed. The missing values may be caused by equipment malfunctions or data storage problems that prevent data from being captured. In data extraction and monitoring applications, missing data and outliers are frequent issues. Tasks like correcting or deleting outliers and filling in NaN values are performed in pre-processing in this work.

Median imputation is a statistical method for handling missing data in the dataset. Its median value replaces a dataset's missing values in this method. Since the median is the centre value of the dataset when arranged from least to most significant, it is an excellent indicator of central tendency,

mainly when there are outliers. Unlike mean imputation, which extreme values can heavily influence, median imputation provides a more robust estimate that better reflects the central tendency of the data without being skewed by outliers. It is less likely that the overall distribution or variance of the data will be distorted when the median value is used for imputation, which ensures that the data is still intact for further analysis (Ratolojanahary et al. 2019). Given the limited extent of missing data, its impact on the final results is expected to be minimal; however, accurate imputation ensures that the integrity of temporal and spatial trends remains intact. Addressing this small yet systematic gap is critical to maintaining the reliability of ML predictions, as even minor inconsistencies in data quality can propagate errors in predictive modeling.

WQI calculation

The Water Quality Index (WQI) is calculated using the Weighted Index Method, which assigns a weight to each water quality parameter based on its significance in assessing overall water health (Talpur et al. 2020).

This study determines the WQI through a structured methodology involving parameter selection, standardization, weight assignment, sub-index computation, and final aggregation (Cude 2001). Several water quality investigators have used the arithmetic weighted index method. Weighted sub-index values are combined to create a final WQI score, comprehensively measuring the water body's health (Bhutiani et al. 2014). The first step in this process is the selection of parameters that are critical in determining water quality. This study has parameters including temperature, pH, BOD, Total coliform, DO, chlorides, TDS, calcium, and hardness. These parameters represent different aspects of water quality, including chemical composition, microbial contamination, and overall suitability for consumption or aquatic life.

Once the parameters are selected, the next step is standardising parameter values to ensure consistency. Since each parameter is measured in different units, they must be converted into a common scale by comparing them with standard permissible limits defined by regulatory agencies such as WHO, BIS, or CPCB. This helps normalize the impact of

each parameter on the final index. The standard values have been considered as per the Bureau of Indian Standards (BIS: IS 10500:201) in this study (Meher et al. 2015.; Singh et al. 2021). Based on the weighting factor in Table 2, different water quality components were multiplied by a simple arithmetic mean to obtain a quality rating. Equation 1 was used to calculate the quality rating scale (Q_i) for each parameter in the dataset used in this study;

$$Q_i = \left\{ \frac{(V_{actual} - V_{ideal})}{(V_{standard} - V_{ideal})} * 100 \right\} \quad (1)$$

For each n water quality parameter, Q_i represents its quality rating, V_{actual} represents its actual value derived from laboratory analysis, V_{ideal} represents its ideal value sourced, and $V_{standard}$ is the parameter's value recommended by water quality guidelines.

Following this, weights are assigned to each parameter based on its relative significance in influencing water quality. Parameters significantly impacting human health and environmental stability, such as DO, BOD, and Total Coliform, are given higher weights. In comparison, less critical parameters receive lower weights. Generally, a lower WQI indicates better water quality, while a higher WQI indicates poorer water quality. In addition to facilitating easy comparison and interpretation, this standardized index can also monitor and manage water resources more effectively (Abba et al. 2024). Table 2, given below, presents the respective weights and standards of various parameters used for WQI computation (Matta et al. 2021).

After determining the weights, sub-indices are calculated for each parameter by multiplying their standardized values with their respective weights. This step quantifies the contribution of each parameter to the overall WQI. Finally, the sub-indices are summed up to obtain the overall WQI score using Eq. 2:

$$WQI = \frac{\sum W_i * Q_i}{\sum W_i} \quad (2)$$

Table 2 Parameter Weightage for WQI Computation

Parameter	Standards	Weightage(W_i)
pH	6.5 to 8.5	0.150
BOD	≤ 3 mg/L	0.225
Total Coliform	≤ 500 MPN/100 mL	0.150
Temperature	≤ 30 °C	0.05
DO	≥ 5 mg/L	0.225
Chlorides	≤ 250 mg/L	0.05
Calcium	≤ 75 mg/L	0.03
Hardness	≤ 200 mg/L	0.10
TDS	≤ 500 mg/L	0.02

Where Q_i is the quality rating, and W_i is the relative weighting.

The study provided a holistic assessment of water quality variations across different monitoring sites by assigning appropriate weights to each parameter based on its significance. The computed WQI values, ranging between 90 and 110 with an average of 96, indicate periods of moderate pollution, necessitating continuous monitoring and proactive interventions. Following the computation of the WQI, ML-based models were applied to predict water quality trends, demonstrating the potential of advanced predictive analytics in environmental management.

Model implementation

For model implementation, the dataset was split into training and testing sets using an 80:20 ratio, where 80% of the data was used for training the ML models and 20% was reserved for testing and validation to evaluate model performance (Kouadri et al. 2021). This study's software used for data analysis was primarily Python, a widely utilized programming language for ML and data science. Key libraries included Scikit-learn for implementing ML models such as XGBoost Regressor (XGBR), Random Forest Regressor (RFR), and Support Vector Regressor (SVR), and TensorFlow and Keras for building and training the Recurrent Neural Network (RNN) model. Data preprocessing, including handling missing values, scaling, and splitting the dataset, was performed using Pandas and NumPy, while Matplotlib and Seaborn were used for visualizing data trends and model performance. The following models have been used for predicting WQI:

XGB regression

A popular ML-based model for regression and classification is Extreme Gradient Boosting (XGB), which is scalable and reliable. The algorithm is a member of the boosting algorithm class, which takes weak predictions and turns them into strong forecasts. XGB commonly uses decision trees as weak learners. Boosting Framework: XGB builds models one after the other. The new models try to fix the mistakes that the older ones committed. It is achieved by concentrating on the previous iteration's residual mistakes. The difference between the expected and actual numbers is minimized to determine the goal function. The Mean Square Error (MSE) is a standard measure among regression loss functions (Uddin et al. 2022). A definition of L in XGB is as follows in Eq. 3:

$$L = \sum_{i=1}^n (y_i, \hat{y}_i) + \sum_{j=1}^{K\Omega} (f_j) \quad (3)$$

The total objective function combines the prediction loss and regularization.

Where,

$L, y_i, \hat{y}_i, \Omega, f_j, n, K, \sum$ j-th decision tree in the ensemble, the number of data points, the total number of trees in the ensemble, and summation over all data points (i) or trees (j), respectively.

XGB's regularization term penalizes model complexity and prevents overfitting. XGB minimizes the loss function by using gradient descent. It calculates the gradient of the loss function for the predicted values for each iteration. It combines the benefits of boosting and regularization to enhance performance and prevent overfitting. Due to its ability to handle large datasets and complex interactions among features, it is a popular choice for regression analyses.

RF regression

When using RFs, several decision trees are built during training, and the output for regression tasks is the average of their predictions. By combining many trees, this technique lowers overfitting and increases prediction accuracy (Kulisz et al. 2021). It works in the following stages:

- I. Bootstrap Aggregating (Bagging): RF creates multiple subsets of training data with random sampling and replacement using bagging.
- II. Decision Trees: Each tree is trained on a different data set and makes predictions. Averaging the predictions from many trees reduces variance and improves generalization by reducing overfitting from individual trees.
- III. Splitting Nodes: RF chooses a random subset of features at each split for each decision tree to build. As a result of this randomness, the trees are decorrelated, and the ensemble is more robust.
- IV. Aggregation: The final prediction is the average of all predictions for regression. The RF regression prediction is given by Eq. 4:

$$\hat{y}_i = \frac{1}{M} \sum_{j=1}^M \hat{y}_i^j \quad (4)$$

Where,

$\hat{y}_i, \hat{y}_i^j, M$ Are the Predicted value, Prediction from the i-th decision tree, and Number of trees in the forest respectively.

n_estimators: Number of trees in the forest.

max_depth: Maximum depth of the trees.

min_samples_split: Minimum number of samples required to split an internal node.

With RF Regression, accurate predictions can be made while controlling for overfitting by taking advantage of the strength of multiple decision trees. In addition to being robust to noise, it can handle both numerical and categorical features efficiently.

SVM regression

Support Vector Machine (SVM) Regression, or Support Vector Regression (SVR), is a type of regression algorithm that finds a function with a margin of tolerance (epsilon) that deviates from the actual observed values while also ensuring that the function is as flat as possible (Li et al. 2017).

This method finds a function $f(x)$ that remains within a specified margin epsilon of the actual values. Optimum flatness is achieved by minimizing the coefficients' norm. In SVR, the loss function is epsilon-insensitive, penalizing deviations larger than epsilon only. Based on a given data point, the loss function is shown as follows in Eq. 5:

$$L \in (y, \hat{y}) = \begin{cases} 0 \rightarrow \text{if } |y - \hat{y}| \leq \epsilon \\ |y - \hat{y}| - \epsilon \rightarrow \text{otherwise} \end{cases} \quad (5)$$

In SVR optimization, the objective function is minimized as follows in Eqs. 6 and 7:

$$\frac{1}{2} ||w||^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (6)$$

$$By_i - (w^T x_i + b) \leq \epsilon + \xi \quad (7)$$

Where,

$L, y, \hat{y}, \epsilon, w, ||w||^2, C, \xi, \xi^*, x_i, b$ Are the loss function measuring the difference between the actual value (y) and the predicted value, The actual target value, The expected value from the model, The margin of tolerance where no penalty is applied if the prediction error is within this range, The weight vector defining the hyperplane in feature space, The squared norm of the weight vector, representing model complexity, a regularization parameter controlling the trade-off between minimizing error and model complexity, Slack variables that allow some data points to violate the margin, The i-th input feature vector, the bias term in the hyperplane equation.

Using kernel functions, SVR can transform non-linear problems into a higher-dimensional space where linear regression can be performed. Polynomial and radial basis function (RBF) kernels are common. In addition to being robust to outliers, It can handle high-dimensional data. Using SVR with a margin of tolerance and regularization, it is possible to balance model complexity and training error.

RNN regressor

RNN is used to process sequential data in neural networks. It is beneficial for sequence modeling and time series prediction since these models can keep a concealed state with knowledge about the sequence's earlier pieces. RNNs process sequences and update the hidden state based on the input at hand and the previously learned hidden state by iterating across time steps. According to Eq. 8:

$$h_t = f(W_h h_{t-1} + W_x x_t + b) \quad (8)$$

Where b is a bias term, and f is an activation function (e.g., tanh or ReLU).

Output Layer: For regression tasks, the final output is computed from the hidden state using a linear transformation as depicted by Eq. 9:

$$\hat{y}_t = W_y h_t + b_y \quad (9)$$

Where,

$h_t, f, W_h, h_{t-1}, W_x, x_t, b, \hat{y}_t, W_y, b_y$ respectively represents the hidden state at time t , capturing the current context, while is the hidden state from the previous time step, the input at time t , weight matrices for the last hidden state, and current input. b is the bias term added to the hidden state computation. The activation function f (e.g., tanh or ReLU) introduces non-linearity. The predicted output is next ($\hat{y}$), computed by applying a weight matrix to the hidden state with b as the bias for the output layer.

Number of hidden units: Number of neurons in the hidden layer.

Learning rate: Controls how much the weights are updated.

Batch size: Number of samples processed at each iteration.

Number of epochs: Number of times the model trains over the entire dataset.

Dropout: Prevents overfitting by randomly setting units to zero during training.

Backpropagation through time (BPTT) is usually used for training RNNs. This involves unrolling a network over its sequence length and applying standard backpropagation to compute gradients. Thus, time series forecasting and natural language processing can benefit from RNN Regressions. A hidden state allows RNNs to capture temporal dependencies and make predictions.

In this study, XGBR, RFR, SVR, and RNN are the ML models used for WQI prediction. XGB uses gradient boosting to build highly accurate predictive models by correcting errors in sequential decision trees. RFR, an ensemble method, aggregates multiple decision trees to

enhance prediction stability and reduce overfitting. SVR is a regression technique that fits the best possible line within a threshold, efficiently handling high-dimensional data. RNN, designed for sequential data, captures temporal dependencies, making it ideal for time-series WQI prediction by learning from historical trends. Together, these models provide robust and precise forecasts for environmental monitoring.

Optimization techniques such as regularization, early stopping, and dropout were applied to improve model stability and generalization. Regularization (i.e., L2 for the RNN and XGBR models) helped avoid overfitting, using a penalty on large weights. This was important in RNN training because they trained for a long time, especially when they overtrained on the temporal patterns, and dropout layers were added to alleviate overfitting. In SVR, the regularization + parameter controls the trade-off between low error margin and avoiding overfitting. In conclusion, these optimization strategies resulted in all models performing better than their baseline counterparts, with RNN being particularly effective in capturing temporal trends in the water quality index (WQI). At the same time, XGBR and RFR were both effective for the non-linear relationships in the WQI data, and SVR remained a flexible model for capturing complex relationships.

Performance evaluation

To evaluate how well these models predict the Water Quality Index (WQI), metrics such as R^2 score, RMSE (Root Mean Square Error), MAE (Mean Absolute Error), and MSE (Mean Squared Error) have been employed. Each metric provides distinct yet complementary insights into the model's performance. The coefficient of determination R^2 indicates goodness-of-fit, representing the proportion of variance in WQI that the model successfully explains. MAE measures the average absolute difference between observed and predicted values, making it less sensitive to outliers and presenting results in the same unit as the dependent variable (Kouadri et al. 2021). MSE, on the other hand, calculates the mean of the squared differences between actual and predicted values. A lower MSE signifies better model performance by reflecting closer alignment between predictions and observations. However, MSE is sensitive to outliers due to the squaring of errors, which amplifies more significant deviations. RMSE, derived from the square root of MSE, provides the average magnitude of errors in the same unit as the dependent variable, enabling direct comparison with other models. Its sensitivity to outliers is similar to MSE but remains a popular metric for assessing overall error magnitude (Chai and Draxler 2014). These metrics offer a holistic view of the model's predictive performance.

1. **Mean Absolute Error:** This metric calculates the typical size of the discrepancies between expected and actual values. It averages the absolute discrepancies between the observed and anticipated values for every data point. With lower values denoting higher performance, MAE offers a precise evaluation of prediction accuracy. Equation 10 shows the error and total number of observations (n), which are the difference between the actual value (Y_i) and the anticipated value ($\hat{Y}_i$).

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (10)$$

2. **Mean Square Error:** assesses a forecasting model's accuracy. It measures the average of the squares of the errors or discrepancies between the model's projected values and the observed actual values. Equation 11 shows the error and total number of observations (n), which are the exact value (Y_i) and the anticipated value ($\hat{Y}_i$).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (11)$$

3. **Root Mean Square Error (RMSE):** is a commonly used metric to evaluate a prediction model's accuracy. It calculates the average magnitude of errors between expected and actual values, considering the quantity and direction of errors. The average squared variances between the predicted and observed values are calculated as RMSE. Equation 12 illustrates how the same is expressed mathematically.

$$RMSE = \sqrt{\frac{\sum_{t=1}^T (\hat{y}_t - y_t)^2}{T}} \quad (12)$$

4. **R² Score:** The variation percentage is determined when independent factors impact the dependent variable. The determination coefficient (K-Score) is more precise and instructive, with no interpretability problems. The determination coefficient also referred to as R² or R², and the percentage of the dependent variable's variation that can be anticipated from the independent variable are shown by R squared, as demonstrated in Eq. 13.

$$R^2 = 1 - \frac{\text{Sum_Squared_Regression(SSR)}}{\text{Total_Sum_of_Squares(SST)}} \quad (13)$$

The sum of the squares of the variations between the actual values that were seen and anticipated values is known as the SSR. The sum of the squares of the variations between the observed absolute values and the observed actual values' mean is known as the SST.

Result and discussion

This section assesses the performance of various models, i.e. XGBR, RFR, SVR, and RNN, on the monthly dataset over six years using data from six different sites using multiple metrics to evaluate these models' performance.

Analysis of water quality parameters and trends

Line graphs depict the rates of change in parameters like pH, BOD, COD, Total Coliform, DO, and TDS over time. Data are analyzed graphically to understand the distribution and trends of various water quality parameters across the six sites over time. As a result of the graphical analysis, key factors influencing water quality trends were identified and evaluated, thus providing valuable insight into the spatial and temporal variation in water quality.

The Water Quality Index (WQI) for Haridwar from 2017 to 2022 across six key sites consistently ranged between 90–110, indicating poor water quality. Seasonal variations show slightly lower WQI values during the monsoon season, suggesting relatively less degradation, and higher WQI values in winter, reflecting further deterioration. Industrial discharges, tourist activities, and seasonal environmental factors influence these trends. During the monsoon (June–September), heavy rainfall dilutes pollutants in the river, resulting in slightly lower WQI values. While surface runoff brings sediments, agricultural chemicals, and urban waste into the river, the increased water flow moderates the overall impact. However, this period also sees a spike in waste generation from religious tourism, such as the Kanwar Yatra, which strains waste disposal systems and contributes to localized pollution hotspots. In winter (December–January), reduced river flow and lower dilution capacity lead to higher WQI values, indicating poorer water quality. Industrial and municipal effluents discharged into the river remain less diluted, and the cold temperatures slow down natural biodegradation processes, exacerbating pollution levels. The influx of tourists during this season, particularly around religious and cultural festivals, further compounds the issue.

An indicator of the pH of a solution is its concentration of hydrogen ions. To ensure the river Ganga in Haridwar remains safe and acceptable, it is crucial to monitor its pH. According to the environmental standards of India, the pH range for drinking water is between 6.5 and 8.5 (Singh et al. 2024). The Ganga at Haridwar is considered acidic if its pH falls below 6.5. Water with an acidic pH can corrode pipes and hand pumps, causing infrastructure damage and releasing harmful metals. Conversely, water with a pH of above 8.5 is classified as alkaline. Alkaline

water may taste bitter or like soda and may not be pleasant to consume. Regular pH monitoring is essential to ensure the safety and taste of Haridwar river water, especially given its cultural, religious, and drinking significance. The analysis in Fig. 3a shows a significant fall in the COVID-19 period for most sites, but for Site 4, this value appears to be significantly high.

Haridwar's Ganga River water quality is assessed based on its biochemical oxygen demand (BOD). BOD measures how much oxygen aerobic microorganisms need to decompose organic matter. When BOD levels are high, there's a lot of biodegradable stuff in the water, which can deplete oxygen levels, adversely affecting aquatic life. According to the environmental standards of India, the permissible BOD level for safe water is usually below 3 mg/L. Maintaining BOD levels within this range is essential to keep the Ganga River at Haridwar healthy and valuable for drinking, bathing, and other things. Elevated BOD levels can result from untreated sewage, industrial effluents, and agricultural runoff, leading to oxygen depletion and potential fish kills. The trend is seen in Fig. 3b. It Shows an increase in BOD values over the last two years and a significant downfall in 2020 (due to COVID-induced lockdown on human activities), with high values indicated from site 5.

Total coliform bacteria are a crucial indicator of water quality for the Ganga River in Haridwar. These bacteria are used to assess the potential presence of pathogenic microorganisms in the water, which can pose serious health risks to humans (Rothmaier et al. 1997). As per the environmental standards of India, the permissible total coliform level in drinking water should be zero per 100 mL. For recreational waters, the acceptable limit is generally below 500 MPN (Most Probable Number) per 100 mL. Monitoring total coliform levels is crucial in Haridwar, where the Ganga River is used for drinking, bathing, and religious activities. High levels of total coliform can indicate contamination from human and animal waste, leading to harmful pathogens such as *E. coli*, which can cause gastrointestinal illnesses and other infections (Chai et al. 2024). Site 5 observed significantly high values, especially in 2019, but these values appear to have reduced for the past three years, as seen in Fig. 3c.

The average temperature of the Ganga River in Haridwar is a significant parameter for assessing its water quality and ecological health. Water temperature influences the river's physical, chemical, and biological processes. According to environmental standards, the average river water temperature should ideally remain within a range that supports aquatic life, typically between 20 °C and 25 °C. In Haridwar, the temperature of the Ganga River can be affected by seasonal variations, industrial discharges, and thermal pollution from nearby activities. Maintaining an appropriate temperature range is essential, as higher temperatures can reduce levels, adversely affecting fish and other aquatic organisms.

Elevated water temperatures can also accelerate the growth of harmful algae and bacteria, leading to deteriorated water quality. Conversely, excessively low temperatures can slow biological processes and affect the river's ecosystem balance. A temperature range of 17–25 °C has been observed, while a higher temperature trend has been observed in the last three years, as seen in Fig. 3d.

DO is a critical parameter for assessing the water quality of the Ganga River in Haridwar. DO levels indicate the amount of oxygen available in the water for aquatic organisms. According to environmental standards, the permissible DO level for maintaining healthy aquatic life is typically above 5 mg/L. High DO levels indicate good water quality, supporting various marine organisms, including fish and macroinvertebrates. Low DO levels, conversely, can be detrimental to aquatic life, leading to hypoxia and potentially causing fish kills and the disruption of marine ecosystems. Factors influencing DO levels include water temperature, flow rate, and the presence of organic matter. Elevated temperatures and excessive organic matter, such as from sewage or agricultural runoff, can deplete oxygen levels. Site 3 has very high levels of average DO among all the sites, as illustrated in Fig. 3e. At the same time, this value seems to be reduced for all the sites in 2022 (enhanced anthropogenic influence at all locations).

There are several human-caused factors contributing to the increase in chloride concentrations nationwide. It denotes contaminated household effluents, connections in septic tanks, and inappropriate sewage disposal or the dumping of solid and animal wastes close to water sources (Yassin et al. 2024). According to environmental standards, the maximum chloride concentration for drinking water is usually less than 250 mg/L. There is a sharp spike in Chloride concentration at Site 2, although the values are below the allowable limits with the rise in recent years, as depicted in Fig. 3f.

Calcium levels are a vital parameter for assessing the water quality of the Ganga River in Haridwar. Calcium, a naturally occurring mineral, is essential for the health of aquatic life and human consumption. According to various environmental standards, calcium concentration in drinking water typically ranges between 30–75 mg/L. However, excessively high calcium levels can lead to water hardness, which can cause scaling in pipes and appliances, affecting water distribution systems and household plumbing. The overall trend shows the values within the limits; however, for site five, these values are high, and for 2021, these values are towards the upper end of the permissible limit, as depicted in Fig. 3g.

TDS is essential for evaluating the Ganga River's water quality in Haridwar. The total amount of all organic and inorganic materials dissolved in water, such as metals, salts, and minerals, is measured by TDS. According to environmental

Fig. 3 Analysis of (a) PH for the study area, (b) BOD for the study area, (c) Total Coliform for the study area, (d) Temperature for the study area, (e) DO for the study area, (f) Chlorides for the study area, (g) Calcium for the study area, (h) TDS for the study area, (i) Hardness for the study area

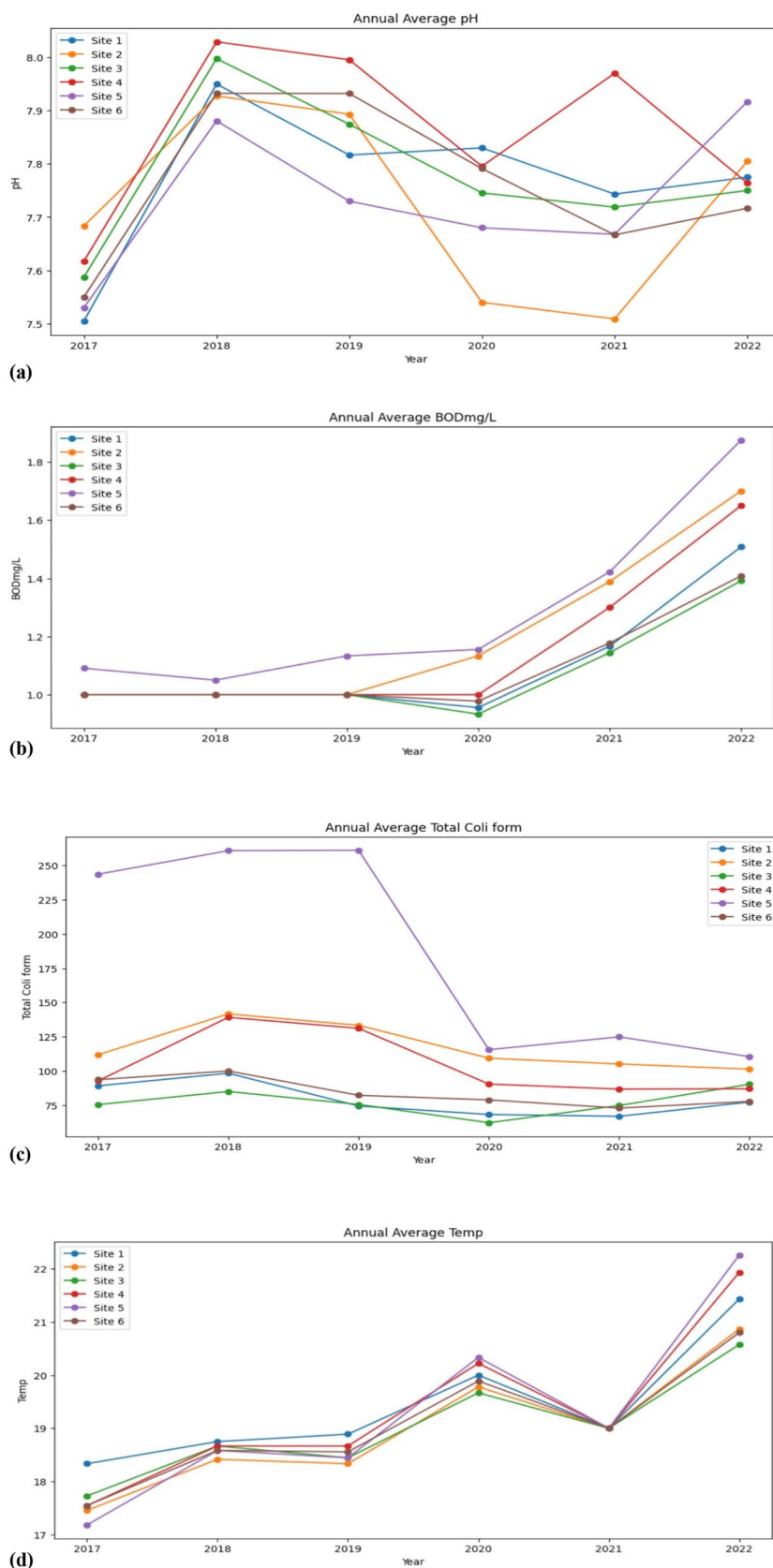
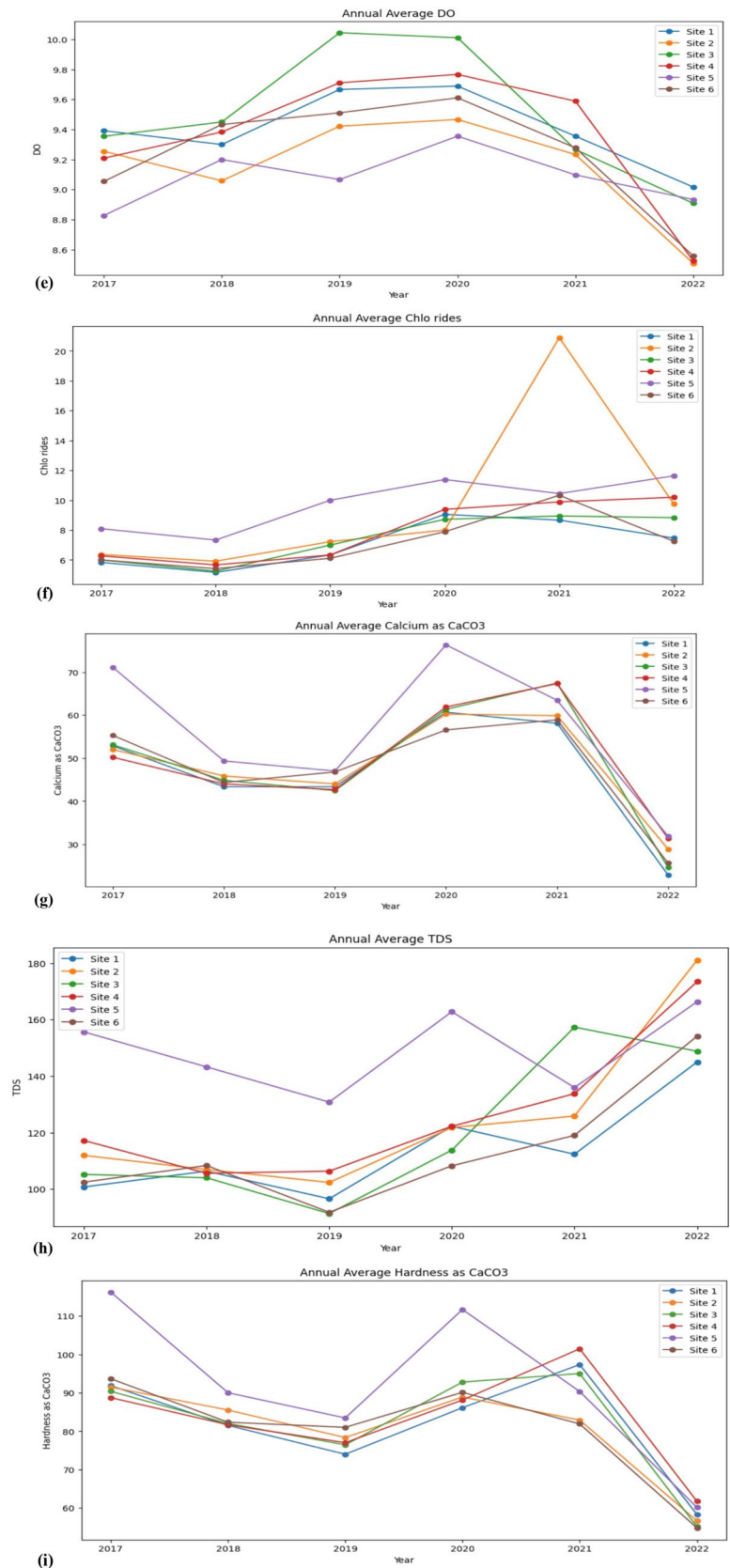


Fig. 3 (continued)



standards, the permissible TDS level for drinking water is typically below 500 mg/L. Monitoring TDS levels is essential in Haridwar, where the Ganga River is used for drinking, bathing, and religious activities. High TDS levels can affect the taste of water, making it unpalatable, and may indicate the presence of harmful contaminants. The values for TDS are below the permissible levels. However, there has been a consistent rise in these values in 2022 (more rock-water interactions in the catchment), as seen in Fig. 3h.

Water hardness is a critical parameter for assessing the quality of the Ganga River in Haridwar. Hardness is primarily caused by dissolved minerals, mainly calcium and magnesium. According to environmental standards, the permissible range for water hardness is typically between 60–120 mg/L as CaCO₃ for drinking water. Maintaining appropriate hardness levels is crucial in Haridwar, where the Ganga River is a significant water source. Too hard water can cause pipes, boilers, and appliances scaling, reducing efficiency and lifespan (Iqbal et al. 2021). Conversely, very soft water can be corrosive, potentially leaching metals from pipes and fixtures. Moderate hardness levels are preferred as they provide essential minerals beneficial for human health and contribute to the stability of aquatic habitats. For site 5, high hardness levels are observed, and compared to other sites, there is a sharp decline for 2022, while the levels have been high for 2020 and 2021.

Further, the M-Kendal test has been performed to understand the overall trend analysis. The WQI trend analysis prediction models indicate that some water quality parameters are significantly related to WQI. DO, TDS, and BOD possess an even higher confidence level slope with p-values above 0.01, meaning they significantly affect WQI. There is also a considerable level of impact from chlorides where the p-value for slope is less than 0.001, indicating a stronger relationship. Total coliform is of moderate significance with a slope greater than 0.5, which shows that coliform's support for WQI changes can be considerable, but these are not the main effects. Parameters categorised as "ns" (not significant) lack statistically meaningful influence on WQI, suggesting their trends do not contribute significantly to the prediction as indicated in Table 3. Overall, this analysis underscores the importance of parameters such as DO, TDS, BOD, and Chlorides as critical indicators for WQI prediction.

Correlation matrices have also been utilized to determine how the attributes are related. The strength or positiveness of the linear relationship (+) between the variables increases with the value's proximity to 1.0. A value of −1.0 signifies a perfect negative linear relationship (−) between the variables, whereas 0.0 suggests no linear relationship. Where the variables move in the same direction, a positive correlation is shown by a positive value; a negative value indicates a negative correlation. As seen in Fig. 4 below, DO had an influential positive association with the WQI, suggesting

Table 3 Trend Analysis using Mann Kendall Test

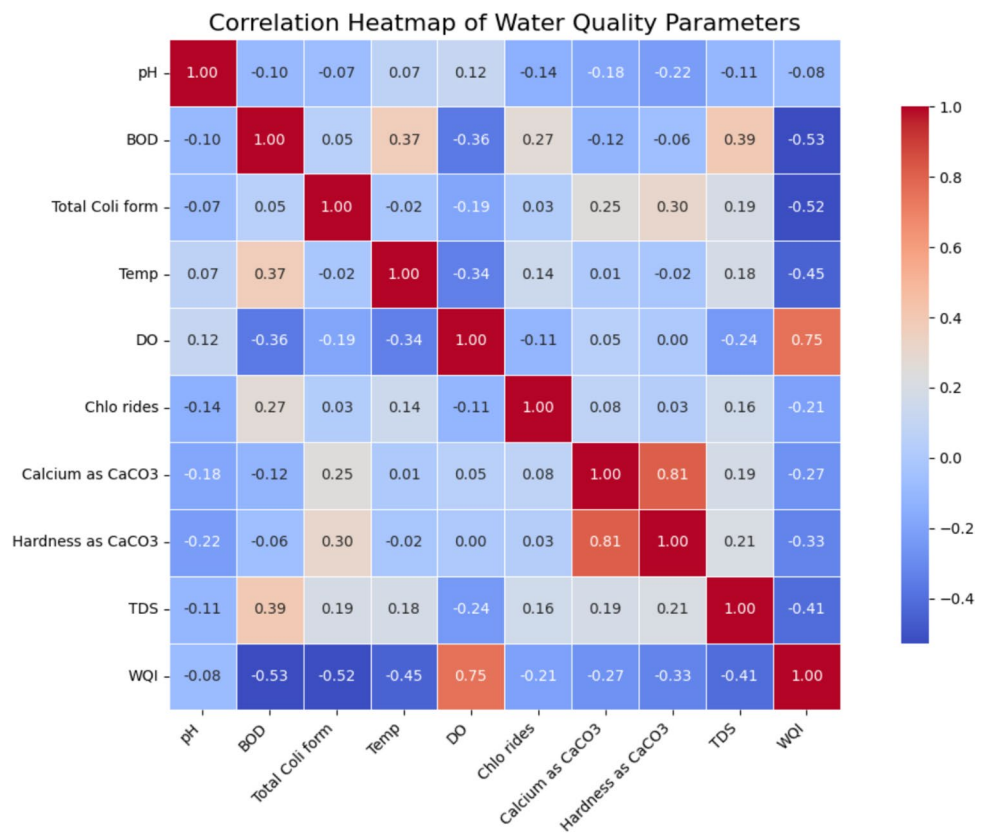
Parameter	Trend	p-value	Slope	Significance
pH	No trend	0.9378	0	ns
BOD	Increasing	0.0011	0	**
Total Coli form	Increasing	0.0187	0.0052494	*
Temp	No trend	0.3293	0	ns
DO	Decreasing	0.0068	− 0.00054795	**
Chlo rides	Increasing	0.0004	0	***
Calcium as CaCO ₃	No trend	0.2397	0	ns
Hardness as CaCO ₃	No trend	0.6342	0	ns
TDS	Increasing	0.0012	0.036765	**
WQI	Decreasing	0.0002	− 0.0060833	***

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns: not significant

that an increase in DO levels will lead to a decrease in WQI (Ganaie et al. 2022). DO is a critical indicator of water health because it supports aquatic life and reflects the ecosystem's self-purification ability. The results of the Mann–Kendall test and the correlation heatmap provide complementary insights into the relationships and trends of parameters influencing WQI. The positive correlation of DO (0.75) with WQI aligns with its importance as a critical indicator of water quality, as higher DO values signify better water conditions. The Mann–Kendall test likely indicates a stable or improving trend for DO over the years, reflecting its central role in sustaining healthy aquatic ecosystems. Its higher correlation value than other parameters suggests that maintaining adequate DO levels should be a priority in water quality management.

On the other hand, BOD has a moderate negative association with the WQI. Total coliform bacteria had a proportionately negative association with WQI, which means an inverse relationship exists between the amount of bacterial contamination and water quality. Increased water temperatures tend to lower water quality, as shown by the moderate negative correlation between temperatures. Parameters like BOD and Total Coliform, with moderate negative correlations, are significant contributors to pollution and should be closely monitored and managed to improve WQI. Temperature and TDS have weaker negative correlations but still indicate that physical and chemical changes in water, often driven by industrial or natural processes, can subtly degrade quality. These results highlight how crucial it is to lower contaminants like BOD and coliform while keeping DO levels high to enhance WQI gradually. This combined trend and correlation analysis provides a strong framework for better monitoring and management of water quality.

These findings highlight the need for stringent industrial and waste management policies, particularly during winter,

Fig. 4 Correlation heat map between various parameters and WQI

and improved waste handling systems during monsoon tourist seasons. Addressing these challenges is critical to mitigating the seasonal impact on the river's water quality and ensuring ecological health.

Analysis of model performance

Monitoring and maintaining water quality parameters in the Ganga River at Haridwar is crucial for ensuring the river's health and usability. Maintaining optimal levels requires effective management practices, including properly treating industrial and domestic wastewater, controlling agricultural runoff, and implementing conservation measures. Regular monitoring involves systematic sampling and analysis of water at various points along the river to detect deviations from the permissible standards. Advanced technologies and methods, such as remote sensing, automated sensors, and laboratory analyses, are employed to measure these parameters accurately. Public awareness and community involvement are also essential in promoting sustainable practices and reducing pollution. By ensuring these parameters remain within safe limits, the quality of the Ganga River can be preserved, protecting its ecological balance, providing safe water for consumption and religious practices, and sustaining its cultural and spiritual significance for the people of Haridwar and beyond.

Table 4 provides an overview of the model's effectiveness in predicting the Ganga River's WQI in Haridwar using a testing dataset.

To forecast the WQI of the Ganga River in Haridwar, this study assessed four ML models, including XGBR, RFR, SVR, and RNN. The models were evaluated using key performance indicators, such as RMSE, MAE, and MSE. These measures quantify the discrepancies between actual and anticipated WQI values and offer a thorough understanding of the model's predictive accuracy. The suggested RNN outperforms these models due to its innate ability to identify temporal connections in WQI data. The RNN gets more robust for this task by processing sequences and learning patterns over time, which helps it recognize trends and seasonal fluctuations. Of the study's alternative models, XGBR is effective with structured data but does not have the temporal modelling needed to

Table 4 Performance Evaluation for WQI Prediction

S No	Model	RMSE	MAE	R ²	MSE
1	XGBR	1.3009	0.9474	0.9128	1.6923
2	RFR	1.4432	0.9844	0.8927	2.0827
3	SVR	1.1609	0.5960	0.9246	0.5866
4	RNN	0.7535	0.5742	0.9655	0.3004

account for WQI's time-dependent character. SVR suffers from scalability and error sensitivity issues in long-span, multi-site datasets yet performs exceptionally well when mapping non-linear relationships. Although resistant to overfitting, RFR performs poorly when dealing with sequential data and tends to perform poorly when data fluctuations are modest.

As seen from Fig. 5, RNNs outperformed all other models, achieving the lowest MAE, MSE, and RMSE. MAE of 0.5742 indicates that the average error in RNN's predictions is minimal, indicating high accuracy. In addition, the MSE coefficient of 0.3004 confirms the effectiveness of this method, and the RMSE coefficient of 0.7535 indicates its superior capability in capturing the temporal dependencies and patterns found in the WQI dataset. Due to the design of the RNN, which includes feedback loops, it can learn from sequential data, making it particularly suitable for time-series predictions as they relate to the WQI.

The SVR model also demonstrated strong performance, with an MAE of 0.5960, indicating a relatively low average prediction error. The MSE of 0.5866, though higher than that of the RNN, still reflects good accuracy, and the RMSE of 1.1609 suggests that SVR effectively handled the WQI data. SVR's robustness to outliers and effectiveness with smaller datasets likely contributed to these favourable results. The XGBR showed competitive performance but was less accurate than the RNN and SVR. The MAE of 0.9474 indicates a higher average prediction error, and the MSE of 1.6923 confirms that the model struggled more with minimizing squared errors. The RMSE of 1.3009 reflects this challenge, suggesting that while it is powerful for handling large

datasets and complex relationships, it may not fully capture the time-dependent characteristics of WQI data.

The RFR was the least effective model in this study, with the highest MAE of 0.9844 and MSE of 2.0827, leading to the highest RMSE value of 1.4432. While RF is typically known for its accuracy and ability to model complex relationships, it is less suited for the intricacies of WQI prediction, mainly when dealing with time-series data. The relatively higher errors suggest that RF had more difficulty capturing the variability and trends in the WQI dataset. Given its superior performance across all criteria, the results show that the RNN model is the most successful in predicting the Ganga River's WQI in Haridwar. The model's architecture allows it to process sequential data efficiently, making it highly suitable for the time-series nature of environmental data like WQI. The SVR also performed well, indicating its robustness in handling WQI prediction, though it fell short of the RNN's accuracy. RFR was the least successful in this situation, and XGBR, despite its strength, demonstrated limitations, underscoring the significance of choosing a model according to the particulars of the dataset.

The R^2 values obtained were 0.9128 for XGBR, 0.8927 for RFR, 0.9246 for SVR, and 0.9655 for RNN. These values indicate that all models demonstrated a strong ability to predict the Ganga River's WQI in Haridwar, with RNN showing the highest R^2 value of 0.9655. This suggests that the RNN model explained 96.55% of the variance in the WQI data, making it the most accurate and reliable model among those tested. The SVR also performed well, with an R^2 of 0.9246, closely followed by XGBR with 0.9128, indicating high predictive power. The RFR, with an R^2 of 0.8927, was the least

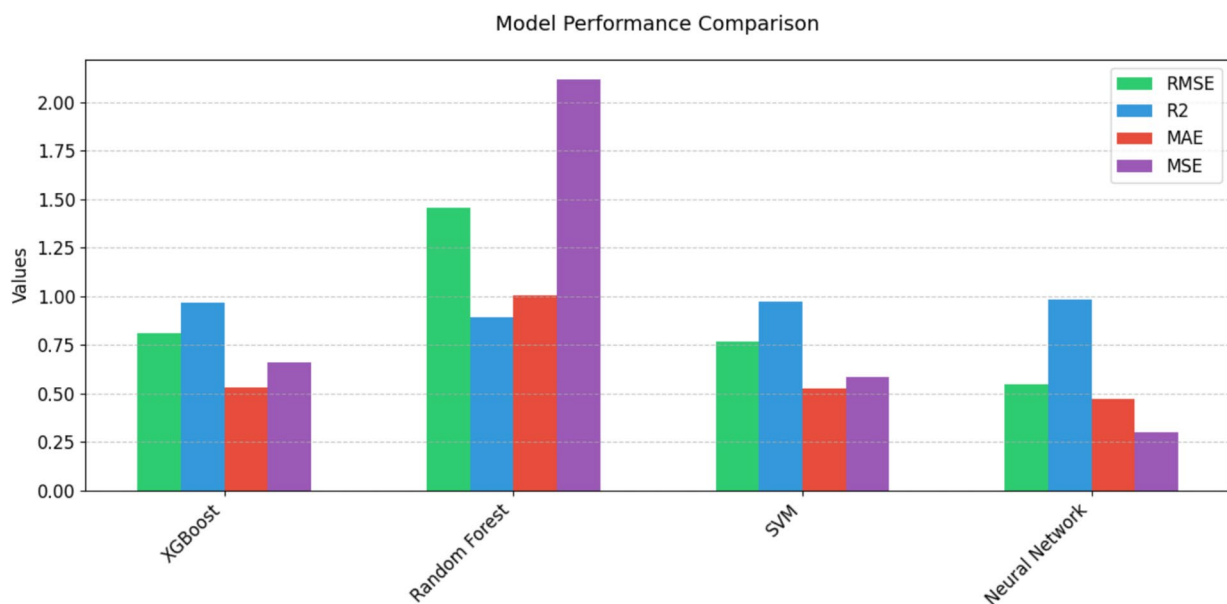


Fig. 5 Performance of various models for WQI prediction

effective in capturing the variance, suggesting slightly lower predictive accuracy than the other models.

These findings underscore the potential of ML in environmental monitoring and management. This nuanced handling of temporal and spatial dynamics makes RNN the most suitable model for WQI prediction, offering unparalleled accuracy and reliability for practical applications. By leveraging advanced models like RNNs, we can achieve higher precision in predicting WQI, enabling more effective and timely interventions to protect and improve the quality of vital water resources like the Ganga River.

Comparative analysis

ML techniques are employed in this proposed work to enhance and refine existing methodologies in existing studies that have also utilized ML to predict the WQI of various bodies. While previous research has demonstrated the potential of these algorithms, such as neural networks, support vector machines, and ensemble models, to improve WQI prediction accuracy, there are still limitations regarding model generalizability and handling data variability. As part of this comparative analysis, Table 5 compares the results against existing studies and identifies the most effective ML models for predicting surface water quality, resulting in more precise and actionable insights into water quality management.

In (Khozani et al. 2022), LSTM is the best-performing model with R^2 : 0.986, RMSE: 1.383, MAE: 0.924, and NSE: 0.992, while in (Setshedi et al. 2021), MLP performs better than ANN with R^2 : 0.9935, and MSE: 39.03. In (Mohseni et al. 2024), XGB outperforms existing models with R^2 : 0.987, RMSE: 3.273, and MAE: 2.727. While R^2 is an essential measure of predictive accuracy, the lower error rates achieved in the proposed work in terms of RMSE, MAE, and MSE (0.7535, 0.5742, and 0.3004) are crucial for improving the reliability of WQI predictions compared to the other studies. In this context, the low RMSE and MAE values are particularly valuable, as they indicate that the proposed model can make more accurate predictions with

more minor deviations, which is essential for environmental monitoring and managing water quality.

Implications and measures

This study sets a precedent for similar global applications to other rivers and water bodies, illustrating how technology is crucial in achieving ecological sustainability. Additionally, it encourages policymakers, environmentalists, and researchers to work together to develop integrated water resource management solutions based on data-driven insights. The following recommendations are suggested for a successful organization in the future:

- Continuous monitoring of critical water quality parameters using automated sensors integrated with machine learning models can provide real-time insights into water quality trends, enabling timely interventions.
- Given the high influx of tourists during the *Kumbh Mela* and other peak seasons, data-driven analysis of waste generation patterns can guide the design and implementation of efficient solid and liquid waste disposal systems.
- Insights from ML models can help prioritize resource allocation, such as deploying cleaning efforts or infrastructure upgrades to sites exhibiting declining water quality trends.
- Authorities should use predictive analytics to identify high-risk areas for pollution and implement targeted mitigation measures, such as stricter industrial discharge regulations or community-led cleanup drives.
- Creating open-access databases for public research. Establishing publicly accessible repositories for water quality data can facilitate collaborative research and innovation, fostering better understanding and solutions for water management challenges.
- Data-driven awareness campaigns and educational programs, underpinned by IEC and BCC strategies, can effectively influence public attitudes toward sustainable water use and waste disposal practices. This is crucial in a city like Haridwar with heavy tourist footfall.

Table 5 Comparative Analysis of the Proposed Work with Existing Studies on WQI Prediction

Parameters	(Khozani et al. 2022)	(Setshedi et al. 2021)	(Mohseni et al. 2024)	Proposed Work
No of features in the dataset	6	8	8	9
Models Used	LSTM , MLP, CNN	ANN, MLP	ANN, SVM, XGB , RF	XGB, SVR, RNN , RFR
Performance Parameters Used	R^2 score, RMSE, MAE, NSE	R^2 score, MSE	R^2 score, RMSE, MAE	RMSE, MAE, MSE, R^2 score
Results	R^2 : 0.986, RMSE: 1.383, MAE: 0.924 and NSE: 0.992	R^2 : 0.9935, MSE: 39.03	R^2 : 0.987, RMSE: 3.273, MAE: 2.727	R^2 : 0.9655, RMSE: 0.7535, MAE: 0.5742, MSE: 0.3004

These suggestions show how advanced technology insights may direct practical, fact-based choices to guarantee sustainable policy planning and water resource management in Haridwar. In addition to providing timely and accurate water quality assessment, this innovative approach promises to revolutionize environmental monitoring. These models can help officials anticipate and handle pollution problems before they become more serious, protecting the health of the river and the welfare of populations that depend on it. Furthermore, this approach can help us comprehend the intricate connection between environmental variables and water quality on a deeper level.

Conclusion

The accurate prediction of the Water Quality Index (WQI) is crucial for addressing environmental pollution and climate change challenges, as rivers serve as vital water sources for agriculture, industry, drinking, and biodiversity. This study demonstrated the potential of machine learning (ML) techniques to predict the WQI of the Ganga River in Haridwar using key water quality indicators such as pH, DO, BOD, COD, Total Coliform, TDS, Hardness, Alkalinity, Calcium, and Magnesium. Various ML models, including XGBoost Regressor (XGB), Random Forest Regressor (RFR), Support Vector Regressor (SVR), and Recurrent Neural Networks (RNN), were applied for WQI prediction. Among them, the RNN model achieved the best performance, with RMSE: 0.7535, MAE: 0.5742, MSE: 0.3004, and an R^2 score of 0.9655, indicating high accuracy in predicting water quality trends. These results confirm that ML models can effectively analyze complex water quality patterns and provide proactive pollution mitigation strategies. However, the reliance on a limited number of parameters may pose challenges in dynamic environmental conditions.

Future research should incorporate longer time spans, a wider range of geographic locations, and advanced data collection techniques such as satellite imagery, remote sensing, and IoT-based sensors to enhance real-time water quality monitoring. Additionally, interpretable ML models, such as attention-based architectures and explainable AI, can improve both the accuracy and transparency of predictions. Expanding this methodology to other major rivers like the Yamuna, Brahmaputra, and Godavari will provide deeper insights into water quality dynamics across diverse ecosystems. These efforts align with initiatives such as the Namami Gange project and contribute to achieving Sustainable Development Goals (SDGs) 6.3, 6.6, and 11, ensuring the long-term sustainability of riverine ecosystems.

Acknowledgements The authors would like to acknowledge the Uttarakhand Pollution Control Board (UPCB), India, for providing the data.

Author contributions All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by Ishaan Dawar and Jabrinder Singh. The modeling part was performed by Sumita Lamba, and the first draft of the manuscript was written by Ishaan Dawar and Maanas Singal. All authors read and approved the final manuscript.

Funding Not Available.

Data Availability Data will be made available upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

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